

CS 483 - Big Data Mining

Final Project Report

Group Name: Smart Voyage

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Github Link: <https://github.com/syhussein123/SmartVoyage/tree/main>

Problem Introduction

Popular travel destinations change over time due to shifts in tourism demand and climate, and these changes are often reflected in both visitor volume and tourism revenue (Zhou, 2024). The goal of this project is to build a data-driven travel analytics system that combines tourism statistics and climate data to identify long term patterns in destination popularity and spending. Specifically, we aim to:

- Analyze long-term trends in tourism popularity (international arrivals), tourism spending intensity (receipts per arrival), and climate (yearly aggregated temperature).
- Classify destinations as increasing, stable, or decreasing in travel desirability based on their historical tourism trends.
- Cluster countries into groups with similar patterns and project future desirability.

We expect to find interpretable relationships between tourism demand, spending intensity, and climate trends. Our classification models should predict whether a destination is becoming more or less desirable, and clustering should reveal clear groups of countries with similar travel evolution profiles.

Formalization

We formulated this project as a multivariate country year data mining problem, where each observation represents a single country in a specific year with associated tourism and climate columns. The final dataset spans 145 countries from 2000 to 2020, resulting in a structured panel dataset suitable for both predictive modeling and pattern analysis.

Each data point includes both raw features and our engineered variables. Core input features consist of tourism variables such as arrivals, tourism receipts, and GDP, along with climate variables such as average yearly temperature. To enhance predictive performance, we constructed additional features including tourism and receipts growth rates, receipts per arrival (RPA), temperature change, and 5 year trend slopes. A key derived variable in this project is the `trend_score`, which combines tourism demand growth and spending quality into a single metric. This score serves as the foundation for multiple downstream tasks, including classification, forecasting, and ranking.

The unit of analysis is therefore a (country, year) pair, and overall the objective is to extract meaningful insights about how tourism desirability changes over time and across different countries using a combination of supervised and unsupervised learning methods and multiple algorithms.

Algorithms

To analyze global travel desirability, we applied four major data mining algorithms corresponding to classification, clustering, time series forecasting, and ranking. Each algorithm was selected to address a specific aspect of the problem and together they provide a more comprehensive understanding of tourism trends.

Algorithm #1 - Classification Models:

We use supervised classification models to predict whether a country's tourism trend is increasing, stable, or decreasing based on its features. The models we used were Logistic regression, Random Forest, Decision Tree, Gradient Boosted trees (GBT), LinearSVC, and Multilayer Perceptron (MLP).

We chose these models to compare linear vs nonlinear approaches. Logistic Regression serves as a baseline, while the tree based models (random forest, decision trees, GBT) and the neural networks such as MLP capture more complex nonlinear relationships between tourism growth, spending, and climate variables. The goal of this classification algorithm is to accurately classify tourism trends, identify which features are most predictive, and compare model performance across different learning approaches.

Algorithm #2: Ranking System - Desirability Score

We developed a ranking system to evaluate and compare countries based on overall travel desirability. This was implemented using a composite scoring approach that combines multiple tourism and climate indicators into a single score.

The ranking system incorporates features such as trend_score, receipts per arrival, tourism arrivals, tourism receipts, and climate variables. Each feature is weighted to reflect its importance in determining destination desirability. We chose this approach because ranking requires combining multiple dimensions into one interpretable metric. Unlike classification or clustering, this method provides a direct comparison of countries. The goal of this algorithm is to rank the countries from most to least desirable, identify countries that are improving or declining, and provide a clear and interpretable summary of tourism performance.

Algorithm #3: Clustering

For clustering, we applied K-Means to group countries based on similar tourism and climate patterns. Unlike classification, clustering is an unsupervised method that identifies natural groupings without predefined labels. We used features capturing:

- Tourism growth (arrivals_growth, receipts_growth)
- Spending intensity (rpa_growth, receipts per arrival)
- Climate change (temp_change)

To visualize the results, we applied Principal Component Analysis (PCA) to reduce the feature space into two dimensions. This allowed us to plot the clusters and confirm that countries within the same cluster exhibit similar patterns.

Algorithm #4: Time Series Analysis, Trend Estimation, and Forecasting

For time series analysis, we applied linear regression to model how tourism trends change over time for each country. Unlike clustering, this approach focuses on individual countries and captures how their trend score evolves year by year. We utilized historical trend_score values as the target and fit a regression line to estimate overall direction and forecast future values.

To evaluate performance, we used RMSE, which measures the difference between predicted and actual values. This allowed us to assess how well the model captures tourism trends over time.

Experiment

To evaluate our models, we conducted a series of experiments using the cleaned and engineered dataset consisting of 145 countries ranging from 2000 to 2020. The experiments were designed to ensure unbiased evaluation, reliable performance comparison, and meaningful interpretation across all four algorithms.

Classification Experiment

For the classification task, we trained multiple supervised learning models to predict whether a country's tourism trend is increasing, stable, or decreasing.

Before training, we assigned labels using a percentile based approach:

- Bottom 20% of trend scores: decreasing
- Middle 60% of trend scores: stable
- Top 20% of trend scores: increasing

This ensured balanced and meaningful class distributions compared to fixed thresholds which initially had performed poorly since many trend score values were borderline and caused accuracy to be at a low performance (~44% accuracy) for both linear and nonlinear approaches.

We used an 80/20 train-test split, where:

- 80% of the data was used to train the models
- 20% was held out for testing performance

To ensure fair evaluation, we applied a stratified split, preserving the class distribution across both datasets. Models were trained using Spark ML pipelines with feature vectorization and transformation steps. They were all trained using the raw and engineered features from the cleaned dataset

We evaluated classification performance using accuracy, and weighted F1 Score. These metrics ensured both overall correctness and balanced performance across all classes.

Ranking System - Desirability Score Experiment

For the ranking task, we designed a composite scoring experiment to evaluate countries based on overall travel desirability. The approach combines seven key indicators into a single weighted score: (trend_score, receipts per arrival, tourism arrivals, tourism receipts, average temperature, GDP, year-over-year temperature change)

Each indicator was assigned a weight based on its importance. `trend_score` received the highest weight (30%) to maintain consistency with the classification and time series models, as it captures overall tourism momentum. Receipts per arrival was weighted at 20% to reflect spending quality, while tourism arrivals and receipts were each assigned 15% to represent destination scale. Climate-related features and GDP were included with smaller weights to capture environmental and economic context.

Before computing the final score, all features were normalized using min-max scaling to ensure comparability across different scales. Temperature change was inverted during normalization so that higher climate instability negatively impacted the overall score.

Two ranking perspectives were computed from the same dataset. The average ranking aggregates all seven indicators per country across 2000–2020, capturing long-term historical desirability. The recent ranking uses each country's most recent non-COVID year, which was 2019 for 99 out of 145 countries (69%), with 2020 excluded due to pandemic-related distortions in tourism data.

The two rankings were then merged into a combined table, and a rank change metric was derived. Countries were categorized as improving (4 or more positions higher), declining (4 or more positions lower), or stable. This comparison allows us to not only rank destinations globally but also observe whether a country's desirability is gaining or losing momentum over time.

Clustering Experiment

We wanted to get an idea of which countries were similar in terms of tourism and climate patterns. Before clustering, we aggregated the data by country and computed average values for key features. The features we computed are:

- `arrivals_growth`
- `receipts_growth`
- `rpa_growth`
- `temp_change`

This allowed us to ensure that each country was represented by a single feature vector. We then combined these features using Spark ML's vectorization pipeline. K-Means clustering was applied to this vectorized dataset to assign each country to a cluster based on similarity in tourism and climate patterns.

In order to determine the number of clusters, we tested multiple values of k and evaluated them using the silhouette score. Although $k=2$ produced the highest score, it resulted in overly generalized groupings. We selected $k=3$ as it provided a better balance between separation and interpretability.

For visualization, we applied PCA to reduce the multi-dimensional feature space into two components, allowing us to plot the clustered data and verify that countries assigned to the same cluster were positioned closely in the reduced space.

Time Series Experiment

For the time series, we wanted to understand how each country's tourism trend changed over time. We used the cleaned country-year dataset and focused on each country's historical `trend_score` values across years.

We grouped the data by country and created a year index so each country’s first year started at 0. We then used linear regression with the year index as the input and trend_score as the target value. This allowed the model to estimate a trend line for each country.

The slope of the regression line was used to determine whether a country’s tourism trend was increasing or decreasing over time. Positive slopes indicated upward tourism momentum, while negative slopes indicated declining tourism momentum.

In order to effectively evaluate how our model behaved, we utilized RMSE, which measures the difference between predicted and actual trend scores. Lower RMSE values indicated that the regression model fit the country’s historical trend more closely.

We also generated actual vs. predicted plots and forecast trend lines to visually compare how well the model captured each country’s tourism pattern.

Results

Classification Results

The classification models showed significant improvement after refining both the labeling strategy and feature set.

Initially, using fixed threshold labels resulted in poor performance, with Logistic Regression achieving approximately 44% accuracy and Random Forest around 43%. This was due to label ambiguity, where many observations were near classification boundaries. After replacing fixed thresholds with percentile based labels and using key growth features, model performance improved substantially.

The final model results were:

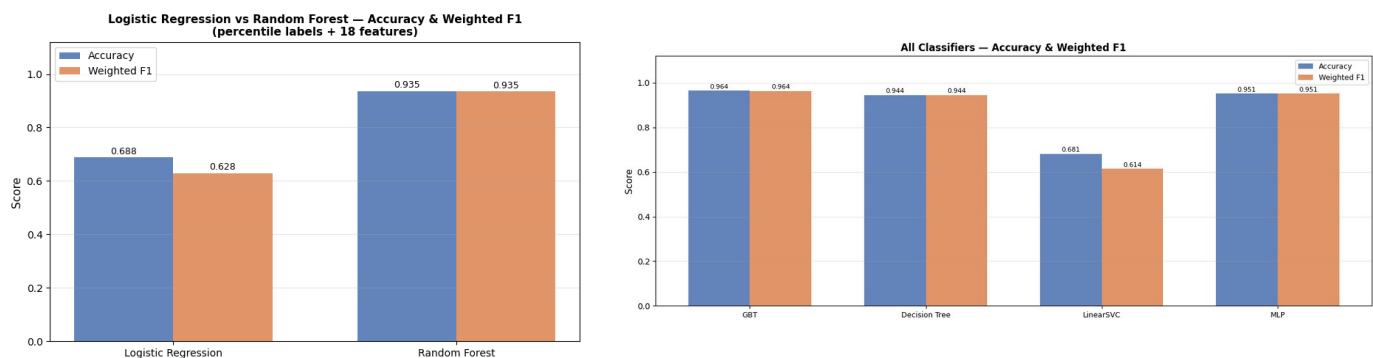


Figure 1: Displays bar charts for the following classification models:

Logistic Regression: ~69% accuracy
Random Forest: ~94% accuracy

Figure 2: Displays bar for the following classification models:

Decision Tree: ~94% accuracy
Gradient Boosted Trees (GBT): ~96% accuracy
Multilayer Perceptron (MLP): ~95% accuracy
LinearSVC: ~68% accuracy

Among all models, Gradient Boosted Trees achieved the highest performance, indicating that nonlinear ensemble methods are best suited for capturing complex relationships in tourism data. The results also show that tree based models outperform linear models, feature engineering significantly improves predictive performance, and proper label design is critical for classification success. Overall, the classification results demonstrate that tourism desirability can be predicted effectively when using appropriate features and models.

Ranking System - Desirability Score Results

The ranking system produced a clear and interpretable ordering of all 145 countries based on overall travel desirability. By combining seven tourism, economic, and climate indicators into a single composite score, we were able to directly compare all countries in a consistent and proportional manner.

Figure 1 shows the top and bottom ranked countries based on the recent desirability score. The separation between high-performing and low-performing destinations is clearly visible. The United States ranked highest with a score of 0.427, followed by Congo (Dem. Rep.) at 0.402 and Qatar at 0.346. These top-ranked countries generally exhibit a strong tourism scale, high spending per visitor, and positive trend momentum. In contrast, the bottom 10 were predominantly cold-climate countries in northern Europe and Central Asia, including Kyrgyz Republic, Kazakhstan, and Tajikistan, which tended to have weaker growth, lower spending intensity, and below-average temperatures that reduced their composite scores.

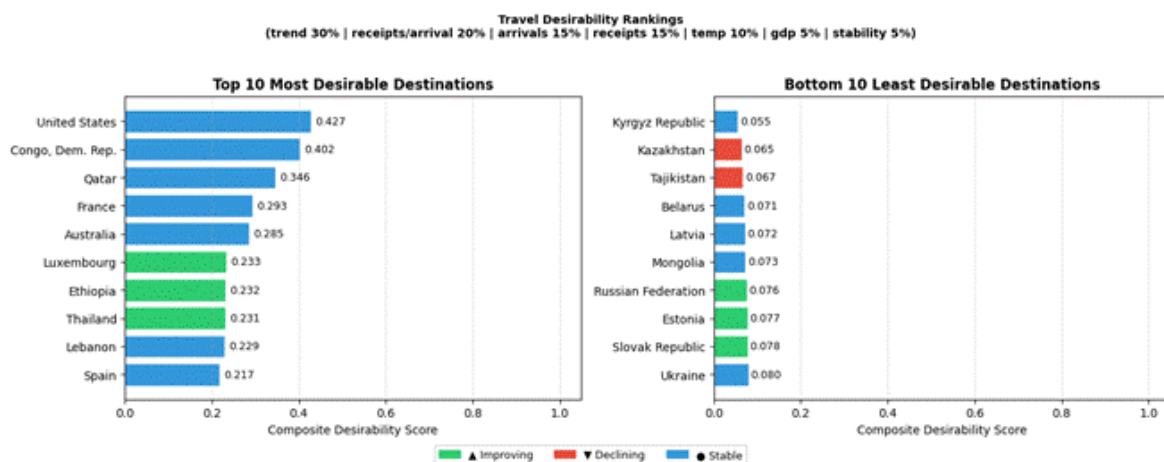


Figure 3: Top 10 and Bottom 10 countries by recent composite desirability score, color-coded by trend direction.

To further quantify these changes, Figure 2 presents the biggest rank changes across the dataset. Poland (+36), Grenada (+34), Austria (+33), and Hungary (+30) also showed strong upward momentum, benefiting from 2019 being a peak year for European tourism. On the declining side, Papua New Guinea fell 80 positions and Bhutan declined 50 positions following policy changes on tourist fees. While a large portion of countries remained stable, a significant number showed noticeable upward or downward

movement, reinforcing the importance of incorporating recent trends alongside long-term averages in the ranking system.

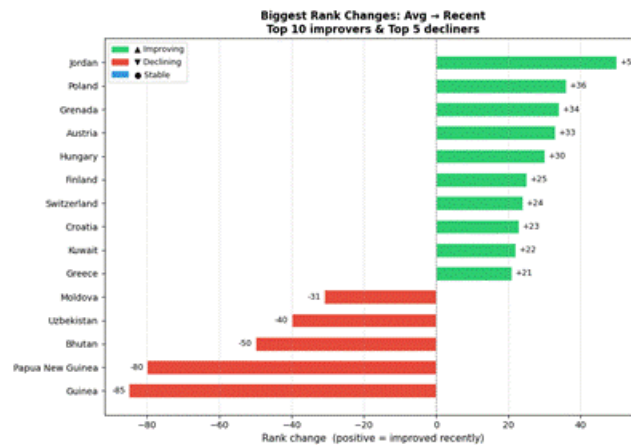
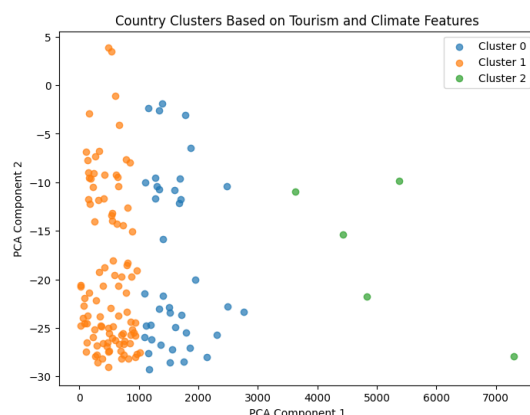


Figure 4: Biggest rank changes from average to recent ranking, showing the top 10 improvers and top 5 decliners.

Overall, the results demonstrate that the composite scoring approach effectively captures both long-term performance and recent momentum, providing a comprehensive and interpretable view of global travel desirability

Clustering Results

K-means clustering revealed clear groupings of countries based on tourism and climate patterns. Using silhouette score as our evaluation metric, we observed that $k=2$ produced the highest score of ~ 0.92 , indicating strong separation. However, this grouping was too broad and did not provide meaningful distinctions between different types of destinations.



We selected $k = 3$, which maintained a high silhouette score (~ 0.80) while producing more interpretable clusters. The PCA visualization further supported this choice, as it showed distinct groupings with clear separation between clusters.

Figure 5: Displays the PCA visual analysis for our clustering algorithm.

The clusters shown in the figure can be interpreted as follows:

- Cluster 0 (blue) contains 40 countries representing moderate or stable destinations with relatively higher spending per visitor, including Indonesia, Thailand, and the Maldives.
- Cluster 1 (orange) contains 100 countries representing fast-growing or emerging destinations, with strong tourism growth but lower spending per visitor, including France, Italy, and the UK.
- Cluster 2 (green) contains only 5 countries representing premium destinations with the highest spending per visitor, including Qatar, Luxembourg, and Australia.

Overall, the clustering results demonstrate that countries naturally group into distinct tourism profiles, highlighting differences in growth, spending behavior, and climate trends.

Time Series Results

The time series model was evaluated using RMSE, which measures the difference between predicted and actual trend scores. Across all 145 countries, the mean RMSE was ~ 0.23 and the median was ~ 0.11 . Since the median is lower than the mean, this suggests that most countries had relatively low prediction error, while a smaller number of countries with irregular tourism patterns produced much larger errors.

South Africa was used as an example country for visualization, with an RMSE of 0.101. In the actual versus predicted plot, most points are reasonably close to the red reference line, indicating that the model captured the general trend well. The forecast plot shows a slight downward trend, suggesting decreasing tourism momentum for South Africa.

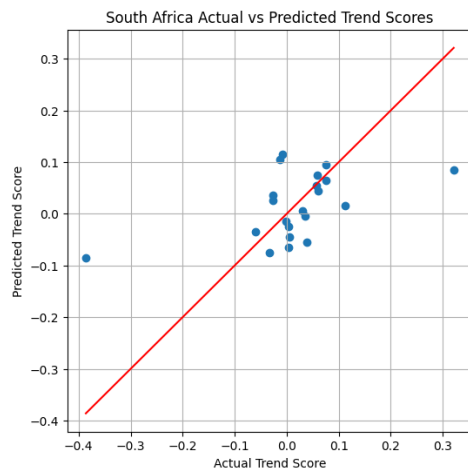


Figure 6: The actual vs predicted plot for our trend scores for South Africa

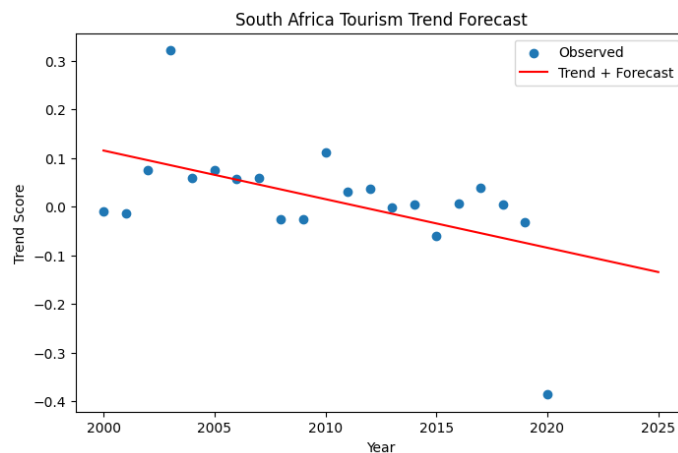


Figure 7: Logistic forecast plot graph for South Africa

The trend slope analysis also identified countries with the strongest increasing and declining tourism momentum. The highest positive trend slope was found for Congo (Dem. Rep.) at 0.578, followed by Libya at 0.249 and Kiribati at 0.212. The strongest declining trend was Sudan at -0.179, followed by Yemen at -0.126 and Tajikistan at -0.096. These slope values help show the direction and strength of each country's tourism trend over time. Below are tables displaying the top ten countries for increasing and declining tourism trends.

Countries with increasing tourism trends		
	country	trend_slope
28	Congo, Dem. Rep.	0.578123
75	Libya	0.248835
69	Kiribati	0.212294
53	Guinea	0.152627
100	Papua New Guinea	0.151055
24	Chad	0.075525
141	Uzbekistan	0.042374
46	Gabon	0.036742
14	Bhutan	0.023678
73	Latvia	0.022389

Table 1: Displays the top ten countries with increasing tourism trends, and their trend slope score.

Countries with declining tourism trends		
	country	trend_slope
123	Sudan	-0.178755
143	Yemen, Rep.	-0.126232
128	Tajikistan	-0.096286
77	Madagascar	-0.095956
61	Iraq	-0.078277
54	Guyana	-0.071786
127	Syrian Arab Republic	-0.054192
119	Solomon Islands	-0.039573
67	Kazakhstan	-0.032760
74	Lebanon	-0.028257

Table 2: Displays the top ten countries with declining tourism trends, and their trend slope score.

The countries with the lowest RMSE values included countries like UAE, Germany, and Switzerland. The results suggest that the linear regression model works best for countries with more stable tourism trends. However, the maximum RMSE was 5.44, indicating that some countries had highly irregular or volatile tourism patterns that were difficult for a linear model to capture. Overall, linear regression provided a useful baseline for trend forecasting, but more complex models may be needed for countries with unstable behavior.

Conclusion

Overall, we created Smart Voyage to be a data driven system analyzing and predicting global travel desirability using tourism and climate data. By combining classification, clustering, time series, and ranking, we were able to capture different aspects of how destinations change over time.

Our Classification model results showed that performance heavily depends on label design and feature engineering. After switching to percentile based labels accuracy improved significantly, with nonlinear models like GBT and random forest performing best. This indicates that tourism trends involve complex, nonlinear relationships.

Clustering helped group countries with similar tourism patterns, while time series analysis allowed us to estimate future trends for individual countries. The ranking system provided an interpretable way to compare destinations using multiple indicators.

This project highlighted how combining multiple data mining techniques provides a more complete and accurate understanding of complex real-world patterns such as global tourism trends.

Limitations:

- Time series used simple linear regression, which struggles with volatile countries
- Forecasting was done per country independently and does not capture relationships between countries
- Data only goes up to 2020 and may not capture sudden global changes
- Feature weights in ranking were manually chosen

Future Work:

- Use more advanced forecasting models such as ARIMA
- Develop multi country forecasting models to capture relationships and shared trends between countries
- Improve automated feature selection and weighting

Overall, this project shows that combining multiple data mining techniques can effectively analyze and predict tourism trends, providing useful insights for decision-making. These results demonstrate the potential of data mining techniques to support data driven decision making in global tourism.

References:

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